

SULIT



SECP2523 DATABASE

SESSION 2024/2025 – SEMESTER 1

ALTERNATIVE ASSESSMENT REPORT: PHASE 2

Name	Lam Yoke Yu
IC No. / Matric No.	040211-08-0642 / A23CS0233
Year / Program	2/SECPH
Section	02
Lecturer Name	Dr. Rozilawati binti Dollah @ Md. Zain
Case Study	Improvement of KKK Online System
System Name	KKK Online System
Group Name	Tech Hi Five

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SECTION C (SQL AND INTERFACE IMPLEMENTATION)

3.1 SQL Statement (DDL & DML)

Add filters and search feature in the salary deduction page

Filter member who has not made salary deduction of the selected month and year (DML)

```
SELECT tb_financial.*, tb_member.m_name
FROM tb_financial
INNER JOIN tb_member ON tb_financial.f_memberNo = tb_member.m_memberNo
LEFT JOIN tb_transaction ON tb_financial.f_memberNo =
tb_transaction.t_memberNo
    AND tb_transaction.t_month = 1
    AND tb_transaction.t_year = 2025
    AND tb_transaction.t_method = 'Potongan Gaji'
WHERE DATE(tb_member.m_approvalDate) < '2025-1-26'
    AND tb_transaction.t_memberNo IS NULL;
```

f_memberNo	f_shareCapital Modal Syer	f_feeCapital Modal Yuran	f_fixedSaving Simpanan Tetap	f_memberFund Tabung Anggota (Al-Abrar)	f_memberSaving Simpanan Anggota	f_dateUpdated	m_name
1004	150	0	0	15	0	2025-01-21 22:52:08	Farra bin Razak
1005	150	0	0	15	0	2025-01-21 22:52:08	Faizul bin Mustafa
1006	150	0	0	15	0	2025-01-21 22:52:08	Razif bin Abdullah
1007	300	0	50	15	0	2025-01-21 22:52:08	Hayati binti Omar
1008	550	0	50	10	0	2025-01-21 22:52:08	Ahmad Zaki bin Mohd Yusof
1009	100	0	0	10	0	2025-01-21 22:52:08	Shazlan bin Sekari @ Shokri
1010	100	0	0	10	0	2025-01-21 22:52:08	Chen Wei Liang
1011	50	0	0	5	0	2025-01-21 22:52:08	Shalini Devi
1012	50	0	0	5	0	2025-01-21 22:52:08	Hasidah binti Ishak

Figure 1 Query Result to Filter member who has not made salary deduction for selected month and year

Search for a specific member (DML)

```
SELECT DISTINCT tb_financial.*, tb_member.m_name
FROM tb_financial
INNER JOIN tb_member ON tb_financial.f_memberNo=tb_member.m_memberNo
WHERE tb_financial.f_memberNo = '1000';
```

f_memberNo	f_shareCapital Modal Syer	f_feeCapital Modal Yuran	f_fixedSaving Simpanan Tetap	f_memberFund Tabung Anggota (Al-Abrar)	f_memberSaving Simpanan Anggota	f_dateUpdated	m_name
1000	410	10	100	25	0	2025-01-23 16:16:45	Roslan bin Roslan

Figure 2 Query Result from Dummy Data to Search for a specific member

Implementation in PHP:

```
SELECT DISTINCT tb_financial.*, tb_member.m_name
FROM tb_financial
INNER JOIN tb_member ON tb_financial.f_memberNo=tb_member.m_memberNo
$where_sql
```

Add fields to upload proof for the additional transactions

Alter table (DDL) or Create table (DDL)

```
ALTER TABLE tb_transaction
ADD COLUMN t_resitNo VARCHAR(255) NOT NULL AFTER t_desc,
ADD COLUMN t_proof VARCHAR(255) NOT NULL AFTER t_resitNo;
```

OR

```
CREATE TABLE tb_transaction (
  t_transactionID INT AUTO_INCREMENT PRIMARY KEY,
  t_transactionType INT NOT NULL,
  t_method VARCHAR(255) NOT NULL,
  t_transactionAmt DOUBLE NOT NULL,
  t_month INT NOT NULL,
  t_year INT NOT NULL,
  t_desc VARCHAR(255) NOT NULL,
  t_resitNo VARCHAR(255) NOT NULL,
  t_proof VARCHAR(255) NOT NULL,
  t_transactionDate timestamp NOT NULL DEFAULT current_timestamp() ON
UPDATE current_timestamp(),
  t_memberNo INT NOT NULL,
  t_adminID INT NOT NULL,
  FOREIGN KEY (t_memberNo) REFERENCES tb_member (m_memberNo),
  FOREIGN KEY (t_adminID) REFERENCES tb_admin (a_adminID),
  FOREIGN KEY (t_month) REFERENCES tb_rmonth (rm_id),
  FOREIGN KEY (t_transactionType) REFERENCES tb_ttype (tt_id)
);
```

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	t_transactionID	int(11)			No	None		AUTO_INCREMENT
2	t_transactionType	int(11)			No	None		
3	t_method	varchar(255)	utf8mb4_general_ci		No	None		
4	t_transactionAmt	double			No	None		
5	t_month	int(11)			No	None		
6	t_year	int(11)			No	None		
7	t_desc	varchar(255)	utf8mb4_general_ci		No	None		
8	t_resitNo	varchar(255)	utf8mb4_general_ci		No	None		
9	t_proof	varchar(255)	utf8mb4_general_ci		No	None		
10	t_transactionDate	timestamp			No	current_timestamp()		ON UPDATE CURRENT_TIMESTAMP()
11	t_memberNo	int(11)			No	None		
12	t_adminID	int(11)			No	None		

Figure 3 Table structure of tb_transaction after executing SQL

Insert new transaction record to the database (DML)

```
INSERT INTO tb_transaction (t_transactionType, t_method, t_transactionAmt,
t_month, t_year, t_desc, t_resitNo, t_proof, t_memberNo, t_adminID)
VALUES (1, 'Transaksi Tambahan', 100, 1, 2025, 'Pembelian Saham',
'R23858', 'transaksi_tambahan_1737604291.png', 1000, '12345678');
```

t_transactionID	t_transactionType	t_method	t_transactionAmt	t_month	t_year	t_desc	t_resitNo	t_proof	t_transactionDate	t_memberNo	t_adminID
76	4	Potongan Gaji	5	12	2024	Potongan Gaji			2025-01-03 22:52:08	1012	12345679
77	1	Transaksi Tambahan	200	1	2025	Pembelian Saham			2025-01-21 22:54:58	1000	12345679
79	1	Transaksi Tambahan	5	1	2025	Cubaan Fail		transaksi_tambahan_1737604291.png	2025-01-23 11:51:31	1000	12345678
80	2	Transaksi Tambahan	10	1	2025	Cubaan Fail		transaksi_tambahan_1737604291.png	2025-01-23 11:51:31	1000	12345678
81	8	Transaksi Tambahan	11436.67	1	2025	Bayaran Balik Bayaran Balik 5		transaksi_tambahan_1737613765.pdf	2025-01-23 14:29:25	1003	12345678
82	3	Potongan Gaji	50	1	2025	Potongan Gaji			2025-01-23 16:16:45	1000	12345678
83	4	Potongan Gaji	5	1	2025	Potongan Gaji			2025-01-23 16:16:45	1000	12345678
84	1	Potongan Gaji	50	1	2025	Potongan Gaji			2025-01-23 16:16:45	1001	12345678
85	4	Potongan Gaji	5	1	2025	Potongan Gaji			2025-01-23 16:16:45	1001	12345678
86	8	Transaksi Tambahan	11436.67	1	2025	Tes: Bayaran Balik 5		transaksi_tambahan_1737630526.jpg	2025-01-23 19:08:46	1003	12345678
87	3	Potongan Gaji	50	1	2025	Potongan Gaji			2025-01-23 19:09:17	1003	12345678
88	4	Potongan Gaji	5	1	2025	Potongan Gaji			2025-01-23 19:09:17	1003	12345678
89	8	Potongan Gaji	91.67	1	2025	Potongan Gaji Bayaran Balik 4			2025-01-23 19:09:17	1003	12345678
90	1	Transaksi Tambahan	50	1	2025	Cubaan Filter		transaksi_tambahan_1737784581.png	2025-01-25 13:56:21	1004	12345678
91	2	Transaksi Tambahan	5	1	2025	No Resit Cubaan	R1573	transaksi_tambahan_1737787965.png	2025-01-25 14:52:45	1003	12345678
92	1	Transaksi Tambahan	100	1	2025	Pembelian Saham	R23858	transaksi_tambahan_1737604291.png	2025-01-26 11:22:34	1000	12345678

Figure 4 Table Content after Executing SQL

Implementation in PHP:

```
INSERT INTO tb_transaction (t_transactionType, t_method, t_transactionAmt,
t_month, t_year, t_desc, t_resitNo, t_proof, t_memberNo, t_adminID)
VALUES ('$type', 'Transaksi Tambahan', '$changeAmount', '$currentMonth',
'$currentYear', '$desc', '$resitNo', '$proofPath', '$memberNo',
'$admin_id');
```

Add fields to the update policies page

Alter table (DDL) or Create Table (DDL)

```
ALTER TABLE tb_policies
DROP p_profitRate,
DROP p_maxFinancingAmt;
```

```
ALTER TABLE tb_policies
ADD COLUMN p_maxAlBai DOUBLE NOT NULL AFTER p_maxInstallmentPeriod,
ADD COLUMN p_rateAlBai DOUBLE NOT NULL AFTER p_maxAlBai,
ADD COLUMN p_maxAlInnah DOUBLE NOT NULL AFTER p_rateAlBai,
ADD COLUMN p_rateAlInnah DOUBLE NOT NULL AFTER p_maxAlInnah,
ADD COLUMN p_maxBPulihKenderaan DOUBLE NOT NULL AFTER p_rateAlInnah,
ADD COLUMN p_rateBPulihKenderaan DOUBLE NOT NULL AFTER
p_maxBPulihKenderaan,
ADD COLUMN p_maxCukaiJalanInsurans DOUBLE NOT NULL AFTER
p_rateBPulihKenderaan,
ADD COLUMN p_rateCukaiJalanInsurans DOUBLE NOT NULL AFTER
p_maxCukaiJalanInsurans,
ADD COLUMN p_maxKhas DOUBLE NOT NULL AFTER p_rateCukaiJalanInsurans,
ADD COLUMN p_rateKhas DOUBLE NOT NULL AFTER p_maxKhas,
ADD COLUMN p_maxKarnivalMusim DOUBLE NOT NULL AFTER p_rateKhas,
ADD COLUMN p_rateKarnivalMusim DOUBLE NOT NULL AFTER p_maxKarnivalMusim,
ADD COLUMN p_maxAlQadrulHassan DOUBLE NOT NULL AFTER p_rateKarnivalMusim,
ADD COLUMN p_rateAlQadrulHassan DOUBLE NOT NULL AFTER p_maxAlQadrulHassan;
```

OR

```
CREATE TABLE tb_policies (
  p_policyID INT AUTO_INCREMENT PRIMARY KEY,
  p_memberRegFee DOUBLE NOT NULL COMMENT 'Fee Masuk',
  p_returningMemberRegFee double NOT NULL,
  p_minShareCapital DOUBLE NOT NULL COMMENT 'Modah Syer Minimum',
  p_minFeeCapital DOUBLE NOT NULL COMMENT 'Modal Yuran Minimum',
  p_minFixedSaving DOUBLE NOT NULL COMMENT 'Wang deposit Anggota Minimum',
  p_minMemberFund DOUBLE NOT NULL COMMENT 'Sumbangan Tabung Kebajikan
Minimum',
  p_minMemberSaving DOUBLE NOT NULL COMMENT 'Simpanan Tetap Minimum',
  p_minOtherFees DOUBLE NOT NULL COMMENT 'Lain-lain',
  p_minShareCapitalForLoan DOUBLE NOT NULL COMMENT 'Modah Syer Minimum',
  p_maxInstallmentPeriod INT NOT NULL COMMENT 'Tempoh Ansuran Maksima',
  p_minShareCapitalForLoan double NOT NULL COMMENT 'Modah Syer Minimum',
  p_maxInstallmentPeriod int(11) NOT NULL COMMENT 'Tempoh Ansuran
Maksima',
  p_maxAlBai double NOT NULL,
  p_rateAlBai double NOT NULL,
  p_maxAlInnah double NOT NULL,
  p_rateAlInnah double NOT NULL,
  p_maxBPulihKenderaan double NOT NULL,
  p_rateBPulihKenderaan double NOT NULL,
```

```

p_maxCukaiJalanInsurans double NOT NULL,
p_rateCukaiJalanInsurans double NOT NULL,
p_maxKhas double NOT NULL,
p_rateKhas double NOT NULL,
p_maxKarnivalMusim double NOT NULL,
p_rateKarnivalMusim double NOT NULL,
p_maxAlQadrulHassan double NOT NULL,
p_rateAlQadrulHassan double NOT NULL,
p_salaryDeductionForSaving DOUBLE NOT NULL COMMENT 'Potongan Gaji
Bulanan untuk Simpanan Tetap',
p_salaryDeductionForMemberFund DOUBLE NOT NULL COMMENT 'Potongan Gaji
Bulanan untuk Tabung Kebajikan',
p_dateUpdated timestamp NOT NULL DEFAULT current_timestamp() ON UPDATE
current_timestamp(),
p_adminID INT NOT NULL,
FOREIGN KEY (p_adminID) REFERENCES tb_admin (a_adminID)
);

```

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	p_policyID 🗝️	int(10)			No	None		AUTO_INCREMENT
2	p_memberRegFee	double			No	None	Fee Masuk	
3	p_returningMemberRegFee	double			No	None		
4	p_minShareCapital	double			No	None	Modah Syer Minimum	
5	p_minFeeCapital	double			No	None	Modal Yuran Minimum	
6	p_minFixedSaving	double			No	None	Wang deposit Anggota Minimum	
7	p_minMemberFund	double			No	None	Sumbangan Tabung Kebajikan Minimum	
8	p_minMemberSaving	double			No	None	Simpanan Tetap Minimum	
9	p_minOtherFees	double			No	None	Lain-lain	
10	p_minShareCapitalForLoan	double			No	None	Modah Syer Minimum	
11	p_maxInstallmentPeriod	int(11)			No	None	Tempoh Ansuran Maksima	
12	p_maxAlBai	double			No	None		
13	p_rateAlBai	double			No	None		
14	p_maxAlInnah	double			No	None		
15	p_rateAlInnah	double			No	None		
15	p_rateAlInnah	double			No	None		
16	p_maxBPulihKenderaan	double			No	None		
17	p_rateBPulihKenderaan	double			No	None		
18	p_maxCukaiJalanInsurans	double			No	None		
19	p_rateCukaiJalanInsurans	double			No	None		
20	p_maxKhas	double			No	None		
21	p_rateKhas	double			No	None		
22	p_maxKarnivalMusim	double			No	None		
23	p_rateKarnivalMusim	double			No	None		
24	p_maxAlQadrulHassan	double			No	None		
25	p_rateAlQadrulHassan	double			No	None		
26	p_salaryDeductionForSaving	double			No	None	Potongan Gaji Bulanan untuk Simpanan Tetap	
27	p_salaryDeductionForMemberFund	double			No	None	Potongan Gaji Bulanan untuk Tabung Kebajikan	
28	p_dateUpdated	timestamp			No	current_timestamp()		ON UPDATE CURRENT_TIMESTAMP()
29	p_adminID 🗝️	int(11)			No	None		

Figure 5 Table structure of tb_policies after executing SQL

Insert new policies to table (DML)

```
INSERT INTO tb_policies (
    p_memberRegFee, p_returningMemberRegFee,
    p_minShareCapital, p_minFeeCapital, p_minFixedSaving,
    p_minMemberFund, p_minMemberSaving, p_minOtherFees,
    p_minShareCapitalForLoan, p_maxInstallmentPeriod,
    p_maxAlBai, p_maxAlInnah, p_maxBPulihKendaraan,
    p_maxCukaiJalanInsurans, p_maxKhas, p_maxKarnivalMusim,
    p_maxAlQadrulHassan,
    p_rateAlBai, p_rateAlInnah, p_rateBPulihKendaraan,
    p_rateCukaiJalanInsurans,
    p_rateKhas, p_rateKarnivalMusim, p_rateAlQadrulHassan,
    p_salaryDeductionForSaving, p_salaryDeductionForMemberFund, p_adminID)
VALUES (
    50, 100,
    300, 50, 5, 0, 0, 0,
    300, 6,
    20000, 20000, 4500, 4500, 10000, 10000, 10000,
    4.2, 4.2, 4.2, 4.2, 4.2, 4.2, 4.2,
    50, 5, 12345678
);
```

p_policyID	p_memberRegFee Fee Masuk	p_returningMemberRegFee	p_minShareCapital Modal Syer Minimum	p_minFeeCapital Modal Yuran Minimum	p_minFixedSaving Wang deposit Anggota Minimum	p_minMemberFund Sumbangan Tabung Kebajikan Minimum	p_minMemberSaving Simpanan Tetap Minimum	p_minOtherFees Lain-lain
1	35	0	300	0	0	5	50	0
2	50	100	300	35	0	5	0	0

p_minShareCapitalForLoan Modal Syer Minimum	p_maxInstallmentPeriod Tempoh Ansuran Maksima	p_maxAlBai	p_rateAlBai	p_maxAlInnah	p_rateAlInnah	p_maxBPulihKendaraan	p_rateBPulihKendaraan	p_maxCukaiJalanInsurans	p_rateCukaiJalanInsurans
300	6	0	0	0	0	0	0	0	0
300	6	20000	4.2	20000	4.2	4500	4.2	4500	4.2

p_maxKhas	p_rateKhas	p_maxKarnivalMusim	p_rateKarnivalMusim	p_maxAlQadrulHassan	p_rateAlQadrulHassan	Potongan Gaji Bulanan untuk Simpanan Tetap	p_salaryDeductionForMemberFund Potongan Gaji Bulanan untuk Tabung Kebajikan	p_dateUpdated	p_adminID
0	0	0	0	0	0	50	5	2023-05-11 00:00:00	12345678
10000	4.2	10000	4.2	20000	4.2	50	5	2025-01-23 21:07:35	12345678

Figure 6 Table Content after Executing SQL

Implementation in PHP:

```
INSERT INTO tb_policies (
    p_memberRegFee, p_returningMemberRegFee,
    p_minShareCapital, p_minFeeCapital, p_minFixedSaving,
    p_minMemberFund, p_minMemberSaving, p_minOtherFees,
    p_minShareCapitalForLoan, p_maxInstallmentPeriod,
    p_maxAlBai, p_maxAlInnah, p_maxBPulihKendaraan,
    p_maxCukaiJalanInsurans, p_maxKhas, p_maxKarnivalMusim,
    p_maxAlQadrulHassan,
    p_rateAlBai, p_rateAlInnah, p_rateBPulihKendaraan,
    p_rateCukaiJalanInsurans,
    p_rateKhas, p_rateKarnivalMusim, p_rateAlQadrulHassan,
    p_salaryDeductionForSaving, p_salaryDeductionForMemberFund, p_adminID)
```



```
VALUES (  
    '$d_memberRegFee', '$d_returningMemberRegFee',  
    '$d_minShareCapital', '$d_minFeeCapital', '$d_minFixedSaving',  
    '$d_minMemberFund', '$d_minMemberSaving', '$d_minOtherFees',  
    '$f_minShareCapitalForLoan', '$f_maxInstallmentPeriod',  
    '$f_maxAlBai', '$f_maxAlInnah', '$f_maxBPulihKenderaan',  
    '$f_maxCukaiJalanInsurans', '$f_maxKhas', '$f_maxKarnivalMusim',  
    '$f_maxAlQadrulHassan',  
    '$f_rateAlBai', '$f_rateAlInnah', '$f_rateBPulihKenderaan',  
    '$f_rateCukaiJalanInsurans', '$f_rateKhas', '$f_rateKarnivalMusim',  
    '$f_rateAlQadrulHassan',  
    '$d_salaryDeductionForSaving', '$d_salaryDeductionForMemberFund',  
    '$admin_id');
```

View the policy changes

Retrieve Policies by Section (DML)

```
SELECT p_memberRegFee, p_returningMemberRegFee,
       p_minShareCapital, p_minFeeCapital, p_minFixedSaving,
       p_minMemberFund, p_minMemberSaving, p_minOtherFees,
       p_adminID, p_dateUpdated
FROM tb_policies
ORDER BY p_policyID ASC;
```

p_memberRegFee Fee Masuk	p_returningMemberRegFee	p_minShareCapital Modal Syter Minimum	p_minFeeCapital Modal Yuran Minimum	p_minFixedSaving Wang deposit Anggota Minimum	p_minMemberFund Sumbangan Tabung Kebajikan Minimum	p_minMemberSaving Simpanan Tetap Minimum	p_minOtherFees Lain-lain	p_adminID	p_dateUpdated
35	0	300	0	0	5	50	0	12345678	2023-05-11 00:00:00
50	100	300	35	0	5	0	0	12345678	2025-01-23 21:07:35

Figure 7 Query Result for Policies for Member Application (Polisi Asas Pemohonan Anggota)

```
SELECT
       p_minShareCapitalForLoan, p_maxInstallmentPeriod,
       p_maxAlBai, p_maxAlInnah, p_maxBPulihKenderaan, p_maxCukaiJalanInsurans,
       p_maxKhas, p_maxKarnivalMusim, p_maxAlQadrulHassan,
       p_rateAlBai, p_rateAlInnah, p_rateBPulihKenderaan,
       p_rateCukaiJalanInsurans,
       p_rateKhas, p_rateKarnivalMusim, p_rateAlQadrulHassan,
       p_adminID, p_dateUpdated
FROM tb_policies
ORDER BY p_policyID ASC;
```

p_minShareCapitalForLoan Modal Syter Minimum	p_maxInstallmentPeriod Tempoh Simpanan Minimum	p_maxAlBai	p_maxAlInnah	p_maxBPulihKenderaan	p_maxCukaiJalanInsurans	p_maxKhas	p_maxKarnivalMusim	p_maxAlQadrulHassan	p_rateAlBai	p_rateAlInnah	p_rateBPulihKenderaan	p_rateCukaiJalanInsurans	p_rateKhas	p_rateKarnivalMusim	p_rateAlQadrulHassan	p_adminID	p_dateUpdated
300	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	12345678	2023-05-11 00:00:00
300	0	20000	20000	4000	4000	10000	10000	20000	4.2	4.2	4.2	4.2	4.2	4.2	4.2	12345678	2025-01-23 21:07:35

Figure 8 Query Result for Policies for Loan Application (Polisi Permohonan Pembiayaan)

```
SELECT p_salaryDeductionForSaving, p_salaryDeductionForMemberFund,
       p_adminID, p_dateUpdated
FROM tb_policies
ORDER BY p_policyID ASC;
```

p_salaryDeductionForSaving Potongan Gaji Bulanan untuk Simpanan Tetap	p_salaryDeductionForMemberFund Potongan Gaji Bulanan untuk Tabung Kebajikan	p_adminID	p_dateUpdated
50	5	12345678	2023-05-11 00:00:00
50	5	12345678	2025-01-23 21:07:35

Figure 9 Query Result for Policies for Salary Deduction (Polisi Potongan Gaji)

View the transaction history of the members

Retrieve transaction (DML)

```
SELECT t.*, m.m_name, tt.tt_desc, rm.rm_desc
FROM tb_transaction t
LEFT JOIN tb_member m ON t.t_memberNo = m.m_memberNo
LEFT JOIN tb_ttype tt ON t.t_transactionType = tt.tt_id
LEFT JOIN tb_rmonth rm ON t.t_month = rm.rm_id
WHERE t.t_month = 1
AND t.t_year = 2025
AND t.t_memberNo = 1000
ORDER BY t.t_transactionDate DESC
LIMIT 0, 10;
```

t_transactionID	t_transactionType	t_method	t_transactionAmt	t_month	t_year	t_desc	t_resitNo	t_proof	t_transactionDate = 1	t_memberNo	t_adminID	m_name	tt_desc	rm_desc
82	3	Polongan Gaji	50	1	2025	Polongan Gaji			2025-01-23 16:16:45	1000	12345678	Roslan bin Roslan	Simpanan Tetap	Januari
83	4	Polongan Gaji	5	1	2025	Polongan Gaji			2025-01-23 16:16:45	1000	12345678	Roslan bin Roslan	Tabung Anggota	Januari
79	1	Transaksi Tambahan	5	1	2025	Cubaan Fail		transaksi_tambahan_1737604291.png	2025-01-23 11:51:31	1000	12345678	Roslan bin Roslan	Modah Syer	Januari
80	2	Transaksi Tambahan	10	1	2025	Cubaan Fail		transaksi_tambahan_1737604291.png	2025-01-23 11:51:31	1000	12345678	Roslan bin Roslan	Modal Yuran	Januari
77	1	Transaksi Tambahan	200	1	2025	Pembelian Saham			2025-01-21 22:54:58	1000	12345679	Roslan bin Roslan	Modah Syer	Januari

Figure 10 Query Result from Dummy Data

Implementation in PHP:

```
SELECT t.*, m.m_name, tt.tt_desc, rm.rm_desc
FROM tb_transaction t
LEFT JOIN tb_member m ON t.t_memberNo = m.m_memberNo
LEFT JOIN tb_ttype tt ON t.t_transactionType = tt.tt_id
LEFT JOIN tb_rmonth rm ON t.t_month = rm.rm_id
WHERE $where_sql
ORDER BY t.t_transactionDate DESC
LIMIT $start_from, $records_per_page;
```

Count for pagination calculation (DML)

```
SELECT COUNT(*) FROM tb_transaction t
WHERE t.t_month = 1
AND t.t_year = 2025
AND t.t_memberNo = 1000;
```

COUNT(*)
5

Figure 11 Query Result from Dummy Data

Implementation in PHP:

```
SELECT COUNT(*) FROM tb_transaction t WHERE $where_sql;
```

Add more information of admin to the system

Alter table (DDL) or Create table (DDL)

```
ALTER TABLE tb_admin
ADD COLUMN a_name VARCHAR(50) NOT NULL,
ADD COLUMN a_ic VARCHAR(14) NOT NULL,
ADD COLUMN a_email VARCHAR(255) NOT NULL,
ADD COLUMN a_position VARCHAR(50) NOT NULL,
ADD COLUMN a_status BOOLEAN DEFAULT 1,
ADD COLUMN a_registrationDate TIMESTAMP DEFAULT CURRENT_TIMESTAMP;
```

OR

```
CREATE TABLE tb_admin (
  a_adminID INT PRIMARY KEY,
  a_name VARCHAR(50) NOT NULL,
  a_ic VARCHAR(14) NOT NULL,
  a_email VARCHAR(255) NOT NULL,
  a_position VARCHAR(50) NOT NULL,
  a_status BOOLEAN DEFAULT 1,
  a_registrationDate TIMESTAMP DEFAULT CURRENT_TIMESTAMP;
);
```

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	a_adminID 	int(11)			No	None		
2	a_name	varchar(50)	utf8mb4_general_ci		No	None		
3	a_ic	varchar(14)	utf8mb4_general_ci		No	None		
4	a_email	varchar(255)	utf8mb4_general_ci		No	None		
5	a_position	varchar(50)	utf8mb4_general_ci		No	None		
6	a_status	tinyint(1)			Yes	1		
7	a_registrationDate	timestamp			No	current_timestamp()		

Figure 12 Table structure of tb_admin after executing SQL

Register new admins to the system

Insert into tb_admin (DML)

```
INSERT INTO tb_admin(a_adminID, a_name, a_ic, a_email, a_position)
VALUES (12345670, 'Lam', '010101-01-0101', 'lam@gmail.com', 'Pengurus');
```

a_adminID	a_name	a_ic	a_email	a_position	a_status	a_registrationDate
12345670	Lam	010101-01-0101	lam@gmail.com	Pengurus	1	2025-01-26 00:39:43
12345678					1	2025-01-24 23:13:52
12345679					1	2025-01-24 23:13:52

Figure 13 Table after executing SQL

Implementation in PHP:

```
INSERT INTO tb_admin(a_adminID, a_name, a_ic, a_email, a_position)
VALUES ('$adminID', '$name', '$ic', '$email', '$position');
```

Insert into tb_user (DML)

```
INSERT INTO tb_user(u_id, u_pwd, u_type)
VALUES (12345670,
'$2y$10$RHYsPeOpkHJOz1Jv4/41..YE.2WERrzHhATrZgbp7FyYLCmikavye', 1);
```

u_id	reset_token	token_expiry	u_pwd	u_type
1000	fbedd7beb60fab5063a7da5bae7b450b	2025-01-21 16:48:00	\$argon2id\$v=19\$m=16,t=2,p=1\$TTI5eHU4TWpCWXd1Tllsdg...	2
1001		2025-01-21 16:35:25	\$argon2id\$v=19\$m=16,t=2,p=1\$TTI5eHU4TWpCWXd1Tllsdg...	2
1003		2025-01-21 16:35:32	\$argon2id\$v=19\$m=16,t=2,p=1\$TTI5eHU4TWpCWXd1Tllsdg...	2
1004		2025-01-21 21:48:44	\$argon2id\$v=19\$m=16,t=2,p=1\$TTI5eHU4TWpCWXd1Tllsdg...	2
1005		2025-01-21 21:48:48	\$argon2id\$v=19\$m=16,t=2,p=1\$TTI5eHU4TWpCWXd1Tllsdg...	2
1006		2025-01-21 21:48:52	\$argon2id\$v=19\$m=16,t=2,p=1\$TTI5eHU4TWpCWXd1Tllsdg...	2
1007		2025-01-21 21:48:56	\$argon2id\$v=19\$m=16,t=2,p=1\$TTI5eHU4TWpCWXd1Tllsdg...	2
1008		2025-01-21 21:48:59	\$argon2id\$v=19\$m=16,t=2,p=1\$TTI5eHU4TWpCWXd1Tllsdg...	2
1009		2025-01-21 21:49:02	\$argon2id\$v=19\$m=16,t=2,p=1\$TTI5eHU4TWpCWXd1Tllsdg...	2
1010		2025-01-21 21:49:04	\$argon2id\$v=19\$m=16,t=2,p=1\$TTI5eHU4TWpCWXd1Tllsdg...	2
1011		2025-01-21 22:19:01	\$2y\$10\$ZIXf7EN/CN53U4PWRFHlHez1lsyj/4J.8OmBqDDeV0A...	2
1012		2025-01-21 22:19:23	\$2y\$10\$RHYsPeOpkHJOz1Jv4/41..YE.2WERrzHhATrZgbp7Fy...	2
12345670		2025-01-26 00:39:43	\$2y\$10\$RHYsPeOpkHJOz1Jv4/41..YE.2WERrzHhATrZgbp7Fy...	1
12345678	077047f56df149ae3bb1970c9df623b2	2025-01-21 14:31:28	\$argon2id\$v=19\$m=16,t=2,p=1\$TTI5eHU4TWpCWXd1Tllsdg...	1
12345679	4b85d76d82c5bcee243b959c7c10fb44	2025-01-21 15:29:23	\$argon2id\$v=19\$m=16,t=2,p=1\$TTI5eHU4TWpCWXd1Tllsdg...	1

Figure 14 Table after executing SQL

Implementation in PHP:

```
INSERT INTO tb_user(u_id, u_pwd, u_type)
VALUES ('$adminID', '$password', 1);
```

3.2 Interface Implementation

Add filters and search feature in the salary



Permohonan ▾ Senarai ▾ Transaksi ▾ Kemaskini Polisi Kemaskini Iklan Laporan

Log Keluar

Transaksi

Anggota yang belum dipotong gaji untuk

Cari Anggota

Pilih Bulan ▾

Pilih Tahun ▾

Tapis

No Anggota

Cari

☐ Select All	No Ahli	Nama	Modah Syer	Modal Yuran	Simpanan Tetap	Tabung Anggota	Simpanan Anggota	Al-Bai	Al-Innah	Baik Pulih Kenderaan	Road Tax dan Insurans	Khas	Karnival Musim Istimewa	Al-Qadrol Hassan	Dikemaskini
☐	1000	Roslan bin Roslan	410.00	10.00	100.00	25.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	23-01-2025
☐	1001	Pravin A/L Ali	250.00	0.00	50.00	25.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	23-01-2025
☐	1003	Lam Kin Xin	300.00	0.00	200.00	25.00	50.00	0.00	0.00	1,833.32	0.00	0.00	0.00	0.00	23-01-2025
☐	1004	Farra bin Razak	200.00	0.00	0.00	15.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	25-01-2025
☐	1005	Faizul bin Mustafa	150.00	0.00	0.00	15.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21-01-2025
☐	1006	Razif bin Abdullah	150.00	0.00	0.00	15.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21-01-2025
☐	1007	Hayati binti Omar	300.00	0.00	50.00	15.00	0.00	0.00	0.00	0.00	0.00	11,898.33	0.00	0.00	21-01-2025
☐	1008	Ahmad Zaki bin Mohd Yusof	550.00	0.00	50.00	10.00	0.00	0.00	0.00	0.00	1,094.72	0.00	0.00	0.00	21-01-2025
☐	1009	Shazlan bin Sekari @ Shokri	100.00	0.00	0.00	10.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21-01-2025
☐	1010	Chen Wei Liang	100.00	0.00	0.00	10.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21-01-2025

1

2

»

Potongan Gaji untuk bulan

Januari ▾

2025 ▾

Potong Gaji

Figure 15 Interface Design for Salary Deduction Page


Log Keluar

Permohonan ▾ Senarai ▾ Transaksi ▾ Kemaskini Polisi Kemaskini Iklan Laporan

Transaksi

Anggota yang belum dipotong gaji untuk

Cari Anggota

Januari ▾ 2025 ▾

Tapis

No Anggota

Cari

☐ Select All	No Ahli	Nama	Modah Syer	Modal Yuran	Simpanan Tetap	Tabung Anggota	Simpanan Anggota	Al-Bai	Al-Innah	Baik Pulih Kenderaan	Road Tax dan Insurans	Khas	Karnival Musim Istimewa	Al-Qadrul Hassan	Dikemaskini
☐	1004	Farra bin Razak	200.00	0.00	0.00	15.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	25-01-2025
☐	1005	Faizul bin Mustafa	150.00	0.00	0.00	15.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21-01-2025
☐	1006	Razif bin Abdullah	150.00	0.00	0.00	15.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21-01-2025
☐	1007	Hayati binti Omar	300.00	0.00	50.00	15.00	0.00	0.00	0.00	0.00	0.00	11,898.33	0.00	0.00	21-01-2025
☐	1008	Ahmad Zaki bin Mohd Yusof	550.00	0.00	50.00	10.00	0.00	0.00	0.00	0.00	1,094.72	0.00	0.00	0.00	21-01-2025
☐	1009	Shazlan bin Sekari @ Shokri	100.00	0.00	0.00	10.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21-01-2025
☐	1010	Chen Wei Liang	100.00	0.00	0.00	10.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21-01-2025
☐	1011	Shalini Devi	50.00	0.00	0.00	5.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21-01-2025
☐	1012	Hasidah binti Ishak	50.00	0.00	0.00	5.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21-01-2025

1

Potongan Gaji untuk bulan

Januari ▾

2025 ▾

Potongan Gaji

Figure 16 Interface Design when filter is applied for month and years


Log Keluar

Permohonan ▾ Senarai ▾ Transaksi ▾ Kemaskini Polisi Kemaskini Iklan Laporan

Transaksi

Anggota yang belum dipotong gaji untuk

Cari Anggota

Pilih Bulan ▾

Pilih Tahun ▾

Tapis

1004

Cari

☐ Select All	No Ahli	Nama	Modah Syer	Modal Yuran	Simpanan Tetap	Tabung Anggota	Simpanan Anggota	Al-Bai	Al-Innah	Baik Pulih Kenderaan	Road Tax dan Insurans	Khas	Karnival Musim Istimewa	Al-Qadrul Hassan	Dikemaskini
☐	1004	Farra bin Razak	200.00	0.00	0.00	15.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	25-01-2025

1

Potongan Gaji untuk bulan

Januari ▾

2025 ▾

Potongan Gaji

Figure 17 Interface Design when filter is applied for member

Add fields to upload proof for the additional transactions



Log Keluar

Permohonan AnggotaPermohonan PinjamanSenarai AnggotaSenarai PeminjamTransaksiKemaskini PolisiKemaskini IklanLaporan

Transaksi Lain

No Anggota

Maklumat Anggota

Name	Roslan bin Roslan
IC	010101-01-0101
No. Anggota	1000
No. PF	1005

Maklumat Saham Ahli

	Status Semasa	Perubahan	Status Baru
Modal Syer	410.00	0.00	410.00
Modal Yuran	10.00	0.00	10.00
Simpanan Tetap	50.00	0.00	50.00
Simpanan Anggota	0.00	0.00	0.00
Tabung Anggota	20.00	0.00	20.00

Jumlah Bayaran: RM 0.00

Ulasan

No. Resit

Bukti Transaksi

Choose File

No file chosen

Hantar

Figure 18 Interface Design for Uploading Proof for Additional Transaction

Add fields to the update policies page



Log Keluar

Permohonan AnggotaPermohonan PinjamanSenarai AnggotaSenarai PeminjamTransaksiKemaskini PolisiKemaskini IklanLaporan

Kemaskini Polisi Permohonan Pembiayaan

Modal Syer Minimum Peminjam

RM

300

.00

Tempoh Ansuran Maksima

6

tahun

Kadar Keuntungan dan Pembiayaan Maksima

Jenis Pinjaman	Pembiayaan Maksima		Kadar Keuntungan	
Al-Bai	RM	20000	.00	4.2 %
Al-Innah	RM	20000	.00	4.2 %
Baik Pulih Kenderaan	RM	4500	.00	4.2 %
Cukai Jalan dan Insurans	RM	4500	.00	4.2 %
Skim Khas	RM	10000	.00	4.2 %
Karnival Musim Istimewa	RM	10000	.00	4.2 %
Al-Qadrul Hassan	RM	10000	.00	4.2 %

Kembali

Kemaskini

Figure 19 Interface Design for Update Policies Page

View the policy changes

 Log Keluar		
Permohonan ▾ Senarai ▾ Transaksi ▾ Kemaskini Polisi Kemaskini Iklan Laporan		
Kemaskini Polisi		
Polisi Asas Pemohonan Anggota		
Perkara	11/05/2023	23/01/2025
Fee Masuk	RM35.00	RM50.00
Fee Masuk Anggota yang Pernah Menjadi Anggota	RM35.00	RM50.00
Modah Syer Minimum	RM300.00	RM300.00
Modal Yuran Minimum	RM0.00	RM35.00
Wang Deposit Anggota Minimum	RM0.00	RM0.00
Sumbangan Tabung Kebajikan Minimum	RM5.00	RM5.00
Simpanan Tetap Minimum	RM50.00	RM0.00
Lain-lain	RM0.00	RM0.00
Admin ID	12345678	12345678
Polisi Permohonan Pembiayaan		
Perkara	11/05/2023	23/01/2025
Modal Syer Minimum Peminjam	300.00	300.00
Tempoh Ansuran Maksima	6 tahun	6 tahun
Pembiayaan Maksima Al-Bai	RM0.00	RM20,000.00
Pembiayaan Maksima Al Innah	RM0.00	RM20,000.00
Pembiayaan Maksima Baik Pulih Kenderaan	RM0.00	RM4,500.00
Pembiayaan Maksima Cukai Jalan dan Insurans	RM0.00	RM4,500.00
Pembiayaan Maksima Skim Khas	RM0.00	RM10,000.00
Pembiayaan Maksima Karnival Musim Istimewa	RM0.00	RM10,000.00
Pembiayaan Maksima Al-Qadrul Hassan	RM0.00	RM20,000.00
Kadar Keuntungan Al-Bai	0.00%	4.20%
Kadar Keuntungan Al-Innah	0.00%	4.20%
Kadar Keuntungan Baik Pulih Kenderaan	0.00%	4.20%
Kadar Keuntungan Cukai Jalan dan Insurans	0.00%	4.20%
Kadar Keuntungan Skim Khas	0.00%	4.20%
Kadar Keuntungan Karnival Musim Istimewa	0.00%	4.20%
Kadar Keuntungan Al-Qadrul Hassan	0.00%	4.20%
Polisi Potongan Gaji		
Perkara	11/05/2023	
Potongan Gaji untuk Simpanan	RM50.00	
Potongan Gaji untuk Tabung Kebajikan	RM5.00	
Admin ID	12345678	

Figure 20 Interface Design for View Policy Changes

View the transaction history of the members

 Log Keluar											
Permohonan ▾ Senarai ▾ Transaksi ▾ Kemaskini Polisi Kemaskini Iklan Laporan											
Sejarah Transaksi											
Pilih Bulan ▾ Pilih Tahun ▾ No Anggota Tapis											
ID	Tarikh	No Anggota	Nama	Jenis	Amaun (RM)	Bulan	Tahun	Ulasan	No Resit	Bukti	No Kerani
91	25-01-2025	1003	Lam Kin Xin	Modal Yuran	5.00	Januari	2025	No Resit Cubaan	R1573	Lihat	12345678
90	25-01-2025	1004	Farra bin Razak	Modah Syer	50.00	Januari	2025	Cubaan Filter		Lihat	12345678
87	23-01-2025	1003	Lam Kin Xin	Simpanan Tetap	50.00	Januari	2025	Potongan Gaji		Tiada bukti	12345678
88	23-01-2025	1003	Lam Kin Xin	Tabung Anggota	5.00	Januari	2025	Potongan Gaji		Tiada bukti	12345678
89	23-01-2025	1003	Lam Kin Xin	Baik Pulih Kenderaan	91.67	Januari	2025	Potongan Gaji Bayaran Balik 4		Tiada bukti	12345678
86	23-01-2025	1003	Lam Kin Xin	Baik Pulih Kenderaan	11,436.67	Januari	2025	Tes: Bayaran Balik 5		Lihat	12345678
82	23-01-2025	1000	Roslan bin Roslan	Simpanan Tetap	50.00	Januari	2025	Potongan Gaji		Tiada bukti	12345678
83	23-01-2025	1000	Roslan bin Roslan	Tabung Anggota	5.00	Januari	2025	Potongan Gaji		Tiada bukti	12345678
84	23-01-2025	1001	Pravin A/L Ali	Modah Syer	50.00	Januari	2025	Potongan Gaji		Tiada bukti	12345678
85	23-01-2025	1001	Pravin A/L Ali	Tabung Anggota	5.00	Januari	2025	Potongan Gaji		Tiada bukti	12345678
81	23-01-2025	1003	Lam Kin Xin	Baik Pulih Kenderaan	11,436.67	Januari	2025	Bayaran Balik: Bayaran Balik 5		Tiada bukti	12345678
79	23-01-2025	1000	Roslan bin Roslan	Modah Syer	5.00	Januari	2025	Cubaan Fail		Tiada bukti	12345678
80	23-01-2025	1000	Roslan bin Roslan	Modal Yuran	10.00	Januari	2025	Cubaan Fail		Tiada bukti	12345678
77	21-01-2025	1000	Roslan bin Roslan	Modah Syer	200.00	Januari	2025	Pembelian Saham		Tiada bukti	12345679
34	21-01-2025	1004	Farra bin Razak	Tabung Anggota	5.00	November	2024	Potongan Gaji		Tiada bukti	12345678
49	03-01-2025	1000	Roslan bin Roslan	Modah Syer	50.00	Disember	2024	Potongan Gaji		Tiada bukti	12345679
50	03-01-2025	1000	Roslan bin Roslan	Tabung Anggota	5.00	Disember	2024	Potongan Gaji		Tiada bukti	12345679
51	03-01-2025	1001	Pravin A/L Ali	Modah Syer	50.00	Disember	2024	Potongan Gaji		Tiada bukti	12345679
52	03-01-2025	1001	Pravin A/L Ali	Tabung Anggota	5.00	Disember	2024	Potongan Gaji		Tiada bukti	12345679
53	03-01-2025	1003	Lam Kin Xin	Simpanan Tetap	50.00	Disember	2024	Potongan Gaji		Tiada bukti	12345679

1 2 3 4 5 »

Figure 21 Transaction History without Filters

 Log Keluar											
Permohonan Senarai Transaksi Kemaskini Polisi Kemaskini Iklan Laporan											
Sejarah Transaksi											
Januari Pilih Tahun No Anggota Tapis											
ID	Tarikh	No Anggota	Nama	Jenis	Amaun (RM)	Bulan	Tahun	Ulasan	No Resit	Bukti	No Kerani
91	25-01-2025	1003	Lam Kin Xin	Modal Yuran	5.00	Januari	2025	No Resit Cubaan	R1573	Lihat	12345678
90	25-01-2025	1004	Farra bin Razak	Modah Syer	50.00	Januari	2025	Cubaan Filter		Lihat	12345678
87	23-01-2025	1003	Lam Kin Xin	Simpanan Tetap	50.00	Januari	2025	Potongan Gaji		Tiada bukti	12345678
88	23-01-2025	1003	Lam Kin Xin	Tabung Anggota	5.00	Januari	2025	Potongan Gaji		Tiada bukti	12345678
89	23-01-2025	1003	Lam Kin Xin	Baik Pulih Kenderaan	91.67	Januari	2025	Potongan Gaji Bayaran Balik 4		Tiada bukti	12345678
86	23-01-2025	1003	Lam Kin Xin	Baik Pulih Kenderaan	11,436.67	Januari	2025	Tes: Bayaran Balik 5		Lihat	12345678
82	23-01-2025	1000	Roslan bin Roslan	Simpanan Tetap	50.00	Januari	2025	Potongan Gaji		Tiada bukti	12345678
83	23-01-2025	1000	Roslan bin Roslan	Tabung Anggota	5.00	Januari	2025	Potongan Gaji		Tiada bukti	12345678
84	23-01-2025	1001	Pravin A/L Ali	Modah Syer	50.00	Januari	2025	Potongan Gaji		Tiada bukti	12345678
85	23-01-2025	1001	Pravin A/L Ali	Tabung Anggota	5.00	Januari	2025	Potongan Gaji		Tiada bukti	12345678
81	23-01-2025	1003	Lam Kin Xin	Baik Pulih Kenderaan	11,436.67	Januari	2025	Bayaran Balik: Bayaran Balik 5		Tiada bukti	12345678
79	23-01-2025	1000	Roslan bin Roslan	Modah Syer	5.00	Januari	2025	Cubaan Fail		Tiada bukti	12345678
80	23-01-2025	1000	Roslan bin Roslan	Modal Yuran	10.00	Januari	2025	Cubaan Fail		Tiada bukti	12345678
77	21-01-2025	1000	Roslan bin Roslan	Modah Syer	200.00	Januari	2025	Pembelian Saham		Tiada bukti	12345679

Figure 22 Transaction History with Month Filter

 Log Keluar											
Permohonan Senarai Transaksi Kemaskini Polisi Kemaskini Iklan Laporan											
Sejarah Transaksi											
Pilih Bulan 2025 1003 Tapis											
ID	Tarikh	No Anggota	Nama	Jenis	Amaun (RM)	Bulan	Tahun	Ulasan	No Resit	Bukti	No Kerani
91	25-01-2025	1003	Lam Kin Xin	Modal Yuran	5.00	Januari	2025	No Resit Cubaan	R1573	Lihat	12345678
87	23-01-2025	1003	Lam Kin Xin	Simpanan Tetap	50.00	Januari	2025	Potongan Gaji		Tiada bukti	12345678
88	23-01-2025	1003	Lam Kin Xin	Tabung Anggota	5.00	Januari	2025	Potongan Gaji		Tiada bukti	12345678
89	23-01-2025	1003	Lam Kin Xin	Baik Pulih Kenderaan	91.67	Januari	2025	Potongan Gaji Bayaran Balik 4		Tiada bukti	12345678
86	23-01-2025	1003	Lam Kin Xin	Baik Pulih Kenderaan	11,436.67	Januari	2025	Tes: Bayaran Balik 5		Lihat	12345678
81	23-01-2025	1003	Lam Kin Xin	Baik Pulih Kenderaan	11,436.67	Januari	2025	Bayaran Balik: Bayaran Balik 5		Tiada bukti	12345678

Figure 23 Transaction with Year and Member Filter

 Permohonan ▾ Senarai ▾ Transaksi ▾ Kemaskini Polisi Kemaskini Iklan Laporan Log Keluar											
Sejarah Transaksi											
Pilih Bulan ▾ 2024 ▾ No Anggota Tapis											
ID	Tarikh	No Anggota	Nama	Jenis	Amaun (RM)	Bulan	Tahun	Ulasan	No Resit	Bukti	No Kerani
34	21-01-2025	1004	Farra bin Razak	Tabung Anggota	5.00	November	2024	Potongan Gaji		Tiada bukti	12345678
49	03-01-2025	1000	Roslan bin Roslan	Modah Syer	50.00	Disember	2024	Potongan Gaji		Tiada bukti	12345679
50	03-01-2025	1000	Roslan bin Roslan	Tabung Anggota	5.00	Disember	2024	Potongan Gaji		Tiada bukti	12345679
51	03-01-2025	1001	Pravin A/L Ali	Modah Syer	50.00	Disember	2024	Potongan Gaji		Tiada bukti	12345679
52	03-01-2025	1001	Pravin A/L Ali	Tabung Anggota	5.00	Disember	2024	Potongan Gaji		Tiada bukti	12345679
53	03-01-2025	1003	Lam Kin Xin	Simpanan Tetap	50.00	Disember	2024	Potongan Gaji		Tiada bukti	12345679
54	03-01-2025	1003	Lam Kin Xin	Tabung Anggota	5.00	Disember	2024	Potongan Gaji		Tiada bukti	12345679
55	03-01-2025	1003	Lam Kin Xin	Baik Pulih Kenderaan	91.67	Disember	2024	Potongan Gaji Bayaran Balik 4		Tiada bukti	12345679
56	03-01-2025	1003	Lam Kin Xin	Baik Pulih Kenderaan	243.33	Disember	2024	Potongan Gaji Bayaran Balik 5		Tiada bukti	12345679
57	03-01-2025	1004	Farra bin Razak	Modah Syer	50.00	Disember	2024	Potongan Gaji		Tiada bukti	12345679
58	03-01-2025	1004	Farra bin Razak	Tabung Anggota	5.00	Disember	2024	Potongan Gaji		Tiada bukti	12345679
59	03-01-2025	1005	Faizul bin Mustafa	Modah Syer	50.00	Disember	2024	Potongan Gaji		Tiada bukti	12345679
60	03-01-2025	1005	Faizul bin Mustafa	Tabung Anggota	5.00	Disember	2024	Potongan Gaji		Tiada bukti	12345679
61	03-01-2025	1006	Razif bin Abdullah	Modah Syer	50.00	Disember	2024	Potongan Gaji		Tiada bukti	12345679
62	03-01-2025	1006	Razif bin Abdullah	Tabung Anggota	5.00	Disember	2024	Potongan Gaji		Tiada bukti	12345679
63	03-01-2025	1007	Hayati binti Omar	Simpanan Tetap	50.00	Disember	2024	Potongan Gaji		Tiada bukti	12345679
64	03-01-2025	1007	Hayati binti Omar	Tabung Anggota	5.00	Disember	2024	Potongan Gaji		Tiada bukti	12345679
65	03-01-2025	1007	Hayati binti Omar	Khas	201.67	Disember	2024	Potongan Gaji Bayaran Balik 6		Tiada bukti	12345679
66	03-01-2025	1008	Ahmad Zaki bin Mohd Yusof	Simpanan Tetap	50.00	Disember	2024	Potongan Gaji		Tiada bukti	12345679
67	03-01-2025	1008	Ahmad Zaki bin Mohd Yusof	Tabung Anggota	5.00	Disember	2024	Potongan Gaji		Tiada bukti	12345679
1 2 3 4 »											

Figure 24 Transaction History with Year Filter

Register new admins to the system



Log Keluar

Permohonan ▾ Senarai ▾ Transaksi ▾ Kemaskini Polisi Kemaskini Iklan Laporan

Daftar Pengurus Sistem

ID Pengurus

Nama

No Kad Pengenalan

Emel

Jawatan

Kata Laluan

Pengesahan Kata Laluan

Hantar

Figure 25 Interface Design to Register New Admin

SECTION D (CONCLUSION AND REFLECTION)

4.1 Conclusion

By adding the filter and search functionality on the salary deduction page and transaction history, the ease of use of the system is increased. Adding a receipt number and file upload for transactions ensures the integrity of the system, preventing the admin from modifying the financial status of a member without an actual transaction. Additionally, adding fields to the update policies page increases the robustness and flexibility of the system.

Adding new pages such as view policy changes and transaction log, can serve an audit purpose to track the changes made by admins. The addition register admin feature ensures that each admin has distinct information and a password, preventing unauthorised access.

However, the newly added register admin feature is still incomplete and not fully functional. It requires refinement to ensure that inputs are validated and passwords are hashed before being stored in the database.

The feature can be further expanded include admin verification and approval before successful registration. In addition, pages to activate and deactivate admins, and edit admin information can also be added. Besides, role-based access control can also be implemented so that different admins have different level of access to different features.

4.2 Reflection

At the beginning of the alternative assessment, I was quite confused about the requirements, as Section A2 says “Based on you module in the WBL” but then Section A3 says “propose a new module.” I was unsure if we shall work on the existing module or create an entirely new one. From my understanding, if we solve the problem within the module, it should be the same module. A new module would only be necessary if a weakness identified in the system (such as a missing feature) required one to complete the system.

I decided to address the weaknesses that I already had in mind while developing the system, comprising both additional features to the existing pages and functionality, and introduce an entirely new feature, that is the admin registration feature.

Next, we shall draw the conceptual and logical ERD. Due to the task distribution in the previous reports for Database course and Software Engineering course, where I focused more on the conceptual ERD with minimal involvement in logical database design. When referring to the P3 Report for Logical Database Design, I realized there were some incorrect cardinalities in some relationships in our logical ERD. Additionally, I realised that the normalisation that was done for P3 is not correct. Since we added an ID for almost all entities, we did not use composite primary key, causing 2nd normal form to be identical to the 1st normal form. I revised this and notified my team members.

For Section 3, I partially completed it before Section 2 because I started making changes to the database and developing after finishing Section 1. In my opinion, there are attributes that I only consider while developing, so it's better to do reverse engineering in this case. However, due to time constraints, I had to proceed with Section 2 before completing Section 3.

For the DDL in Section 3, I was not sure whether we should use ALTER TABLE or CREATE TABLE. Since we are improving on our system and my proposed module does not introduce a

new entity, I thought ALTER TABLE would be more suitable. However, as I refer to the rubric, it is written “All tables are created with complete create table SQL statements with all PKs and FKs are defined correctly,” which implies that I should use CREATE TABLE in this case. I included both CREATE TABLE and ALTER TABLE for the tables that are modified because I wanted to follow the rubric but also do what I felt was correct.

For all the features, since we need to make the interface design, I decided to develop it immediately and integrate them into the WBL project. For the admin registration feature, however, I am still considering whether we should add it to the system, as we need to think about how it should function and integrate with the existing system. When developing for admin registration, I also thought of the problem, if the new admin registers by themselves, we would need add the approval date and the admin who approved the application. If the new admin is registered by an existing admin, a column shall also be added to the admin who aid the action.

Finally, in the conclusion and reflection section, I feel that the reflection is more like a diary entry for the alternative assessment, where I write about my experience throughout the process.